

ONLINE INVENTORY MANAGEMENT SYSTEM

AI23431- WEBTECHNOLOGY AND MOBILE APPLICATION

MINI-PROJECT REPORT

SUBMITTED BY

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Certified that this Project Thesis titled "**ONLINE INVENTORY MANAGEMENT SYSTEM**" is the Bonafide work of LAKSHMI SHRI S (2116241801134) who carried out the work under my supervision. Certified further that to the best of my knowledge the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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INTERNAL EXAMINER

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ABSTRACT

The Online Inventory Management System (SmartIMS) is a comprehensive web-based platform designed to help businesses efficiently track, manage, and analyze their inventory, sales, and purchases. With growing complexities in supply chain and stock management, this system bridges the operational gap by providing a centralized, real-time, and visually interactive dashboard for monitoring stock levels, recording transactions, and generating business reports.

The system provides features for managing product inventory across multiple categories (Electronics, Accessories, Furniture, Stationery), recording sales transactions with customer details, tracking purchase orders from suppliers, and generating analytics reports including total revenue, gross profit, and inventory valuation. It also includes automated low-stock alerts to ensure timely restocking, a monthly sales vs. purchases overview chart, and role-based access control with Admin and User profiles.

Built using modern web technologies including React.js for the frontend with a professional dark-themed UI, the system delivers a seamless and interactive user experience. The application incorporates responsive design principles and features CSV export functionality, interactive charts for data visualization, and a clean dashboard overview showing key business metrics at a glance.

Overall, SmartIMS serves as a digital business tool that empowers inventory managers with real-time insights and data-driven decision-making capabilities, thereby contributing to improved operational efficiency and reduced stock management errors.

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LIST OF ABBREVIATIONS

ABBREVIATION	FULL FORM
IMS	Inventory Management System
UI	User Interface
UX	User Experience
HTML	HyperText Markup Language
CSS	Cascading Style Sheets
JS	JavaScript
API	Application Programming Interface
HTTP	HyperText Transfer Protocol
SPA	Single Page Application
DOM	Document Object Model
CSV	Comma-Separated Values
CRUD	Create, Read, Update, Delete
SKU	Stock Keeping Unit
ERP	Enterprise Resource Planning
KPI	Key Performance Indicator

CHAPTER 1 - INTRODUCTION

1.1 General

Efficient inventory management is a cornerstone of successful business operations across all industries. Organizations dealing with products, supplies, and equipment require robust tools to track stock levels, manage purchases from suppliers, record sales transactions, and generate insightful business analytics. Without proper inventory control, businesses face challenges such as stockouts, overstocking, financial losses, and poor customer satisfaction.

The Online Inventory Management System (SmartIMS) is a web-based digital platform developed to address these operational challenges. The system provides a centralized, real-time inventory dashboard enabling businesses to monitor product stock levels, track low-stock alerts, record and review sales and purchase transactions, and generate comprehensive analytics reports from a single user-friendly web interface.

1.2 Need for Online Inventory Management System

Traditional inventory management using spreadsheets or manual records is error-prone, time-consuming, and lacks real-time visibility into stock status. Small and medium-sized businesses particularly suffer from the absence of an affordable, easy-to-use system that consolidates inventory, sales, and purchasing into one unified platform.

A dedicated online inventory management system bridges this gap by presenting real-time stock data, automated low-stock alerts, category-based filtering, and visual business analytics. SmartIMS supports role-based access (Admin and User), ensuring data security and appropriate access for different team members, along with CSV export for integration with external business tools.

1.3 Scope of the System

The scope of the SmartIMS system encompasses the following areas:

- Managing product inventory with name, category, quantity, unit price, total value, minimum stock level, and real-time status.
- Recording and reviewing sales transactions with product, customer, quantity, and revenue details.
- Tracking purchase orders from suppliers including category, supplier details, quantity, and cost.
- Generating analytics reports showing total revenue, purchases, gross profit, inventory value, top products, and stock by category.

- Providing automated low-stock alerts and a detailed restock report for items needing replenishment.
- Supporting role-based user access with Admin (full access) and User profiles.
- Offering CSV data export functionality for external analysis and reporting.

1.4 Organization of the Report

This report is organized into the following chapters:

- Chapter 1: Introduction - general overview, need, and scope of the system.
- Chapter 2: Literature Survey - existing inventory management systems and related work.
- Chapter 3: Problem Formulation and Objectives of the proposed system.
- Chapter 4: System Design - architecture and system requirements.
- Chapter 5: System Implementation - module-wise description with screenshots.
- Chapter 6: Results and Discussion - performance evaluation and testing.
- Chapter 7: Conclusion and Future Work.

1.5 System Architecture Diagram

The system follows a client-server model where the React.js frontend handles inventory data, sales records, and purchase orders using component-level state management. The architecture supports a responsive dark-themed professional UI for multi-device access and includes role-based authentication at the login layer before granting access to any module.

CHAPTER 2 - LITERATURE SURVEY

2.1 Introduction

Inventory management has evolved significantly with digital transformation. Enterprise Resource Planning (ERP) systems and dedicated inventory management tools have become essential for businesses to maintain competitive operational efficiency. Research in this domain highlights the growing importance of real-time inventory platforms that reduce manual errors and provide actionable analytics. Understanding existing systems helps identify gaps that SmartIMS aims to address.

2.2 Existing Inventory Management Systems

Several platforms have been developed for inventory and stock management. Enterprise-grade systems such as SAP, Oracle NetSuite, and Microsoft Dynamics offer comprehensive ERP solutions but are expensive and complex for small businesses. Mid-tier solutions like Zoho Inventory and Fishbowl provide good functionality but often require significant subscription fees and steep learning curves. Open-source alternatives such as Odoo offer flexibility but require technical expertise to deploy and maintain.

Most existing systems are designed for large enterprises and lack the simplicity required for smaller operations. SmartIMS addresses this gap by providing an intuitive, lightweight web application that handles core inventory operations without the overhead of complex enterprise software.

2.3 Related Work

Research studies on web-based inventory management systems demonstrate that real-time dashboards, automated alerts, and visual analytics significantly improve stock management efficiency. A 2022 study on SME inventory practices highlighted that businesses using digital inventory tools reduced stockout incidents by up to 35% compared to those using manual methods.

Studies on UI design for business applications emphasize the importance of clean dashboards, color-coded status indicators, and interactive charts in enabling rapid decision-making. Systems incorporating low-stock alert mechanisms and automated restock cost estimation have demonstrated measurable improvements in procurement planning efficiency.

2.4 Summary

From the literature survey, it is evident that while several inventory management systems exist, most focus on large enterprise needs. There is significant opportunity for a lightweight yet feature-rich web-based inventory system combining real-time stock tracking with visual analytics and role-based access. SmartIMS is designed to fulfill this need for small and medium-sized businesses.

CHAPTER 3 - PROBLEM FORMULATION AND OBJECTIVES

3.1 Problem Statement

Businesses today face significant challenges in managing inventory without a centralized digital system. Small and medium enterprises often rely on spreadsheets or manual record-keeping, leading to inefficiencies, data errors, and poor visibility into business performance.

Key challenges include:

- Lack of real-time visibility into current stock levels and low-stock situations across product categories.
- Difficulty in tracking sales transactions and identifying top-performing products by revenue.
- Absence of automated low-stock alerts, leading to stockouts and lost sales opportunities.
- No centralized system to manage and review purchase orders from multiple suppliers.
- Difficulty generating comprehensive business analytics reports for revenue, profit, and inventory value.
- Absence of role-based access control, creating data security and accountability risks.

These challenges collectively lead to poor inventory control, financial losses, and inability to make data-driven decisions. A digital inventory management system addresses this critical operational gap.

3.2 Objectives of the System

The primary objectives of the Online Inventory Management System (SmartIMS) are:

- To provide a centralized, real-time web platform for managing product inventory across multiple categories.
- To track stock levels and automatically flag products below their minimum threshold with visual alerts.
- To enable recording and reviewing of sales transactions with details on products, customers, quantities, and revenue.
- To manage and track purchase orders from suppliers with full purchase history visibility.
- To generate comprehensive business analytics reports covering revenue, purchases, gross profit, and inventory valuation.
- To provide visual data representations including bar charts, pie charts, and line graphs for business intelligence.
- To implement role-based user authentication with Admin and User access levels.
- To support CSV export functionality for external reporting and data integration.

CHAPTER 4 - SYSTEM DESIGN

4.1 Introduction

System design defines the structure, components, and interactions of the application. SmartIMS is designed as a modern single-page web application (SPA) delivering a smooth, interactive user experience. The design focuses on data clarity, visual analytics, a responsive dark-themed layout, and ease of navigation across all inventory management functions.

4.2 System Architecture

The system follows a component-based frontend architecture built using React.js. The application is divided into reusable UI components, with React Router handling client-side navigation between Dashboard, Inventory, Sales, Purchases, Reports, and Profile sections. The architecture consists of the following layers:

- **Presentation Layer:** Built with React.js, this layer handles all user interactions, visual charts, and real-time data display using a professional dark-themed UI (SmartIMS v2.0).
- **Authentication Layer:** A login module handles user authentication with role-based credentials (Admin and User) before granting access to the system.
- **Routing Layer:** React Router manages navigation between Dashboard, Inventory Management, Sales Management, Purchase Management, Reports & Analytics, and Profile modules.
- **State Management Layer:** Application state including inventory data, sales records, and purchase orders is managed using React component state and props.
- **Data Layer:** Product inventory, sales transactions, and purchase records are maintained in structured JavaScript data objects supporting full CRUD operations.

4.3 Database Design

SmartIMS uses in-application state management for data storage. The core data entities include: Products (id, name, category, quantity, unitPrice, totalValue, minStock, status), Sales (date, product, customer, qty, unitPrice, total), Purchases (date, product, category, supplier, qty, unitCost, total), and User Profiles (username, role, accessLevel). Future enhancements will incorporate a backend database for persistent multi-user data storage.

4.4 System Requirements

4.4.1 Hardware Requirements

- **Processor:** Intel Core i3 or higher
- **RAM:** Minimum 4 GB
- **Storage:** 256 GB SSD or HDD
- **Internet Connection:** Required for development and deployment

- Input Devices: Keyboard and Mouse

4.4.2 Software Requirements

- Operating System: Windows 10 / macOS / Linux
- Frontend Framework: React.js (v18+)
- Charting Library: Recharts / Chart.js
- Routing: React Router DOM
- Development Environment: Visual Studio Code
- Package Manager: npm / yarn
- Version Control: Git
- Web Browser: Google Chrome / Firefox / Edge

CHAPTER 5 - SYSTEM IMPLEMENTATION

5.1 System Methodology

SmartIMS is developed using an agile, component-based development methodology. The application is built entirely on the frontend using React.js, enabling the creation of reusable, modular UI components. Each section (Dashboard, Inventory, Sales, Purchases, Reports, Profile) is developed as an independent React component with its own state management, promoting maintainability and scalability.

A professional dark-themed UI provides a visually distinct and modern interface suitable for business applications. React Router DOM provides seamless single-page application navigation without full page reloads. Interactive charts are implemented for data visualization in the dashboard and reports sections.

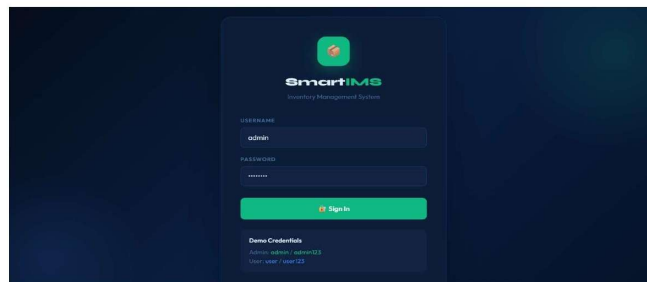
5.2 Module Descriptions

5.2.1 Login / Authentication Module

The Login module is the secure entry point of SmartIMS. It features a clean authentication form with username and password fields against a dark navy-themed background, branded as 'SmartIMS - Inventory Management System'. The module implements role-based authentication supporting two account types: Admin (admin / admin123) with full system access to all modules including inventory CRUD operations, and User (user / user123) with standard read and record access. Demo credentials are prominently displayed on the login page for ease of testing and demonstration. Upon successful authentication, the system routes the user directly to the main Dashboard.

ONLINE INVENTORY MANAGEMENT SYSTEM (PROJECT OUTPUT)

→ LOGIN PAGE



→ DASHBOARD

Figure 5.1: Login Page - SmartIMS Authentication Screen

5.2.2 Dashboard Module

The Dashboard module provides a comprehensive real-time business overview immediately upon login. It is divided into multiple sections displaying key performance indicators (KPIs) and visual analytics. The top navigation bar displays the SmartIMS v2.0 branding, a low-stock warning badge, current time, light/dark mode toggle, and the logged-in user name (Alex Johnson - Admin).

The KPI summary section displays four metric cards: Total Products showing 10 products with \$270k total inventory value, Total Sales showing \$9.7k revenue from 10 transactions, Purchases showing \$28.6k from 5 purchase orders, and a Low Stock Alert card showing 4 items that need immediate restocking.

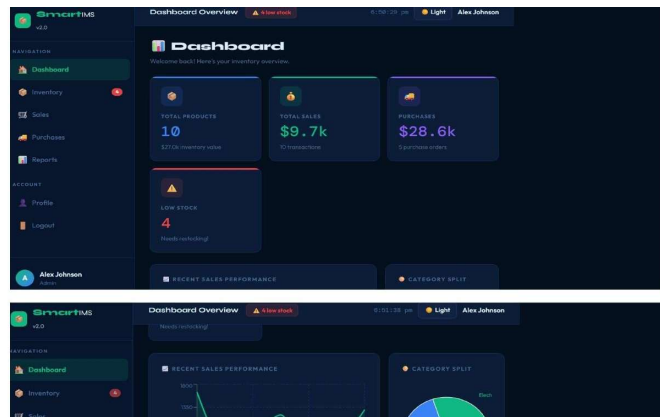


Figure 5.2: Dashboard - KPI Cards and Sales Performance Chart

The dashboard also features a Recent Sales Performance line chart visualizing product-wise revenue trends across categories (4K, Webcam, USB-C, Office, Notebook, Wireless, Laptop), and a Category Split donut pie chart showing inventory distribution. A real-time Low Stock Alerts banner at the bottom prominently identifies the 4 items requiring restocking: Mech Keyboard (6 left), 4K Monitor (4 left), Webcam HD (3 left), and Standing Desk (2 left).

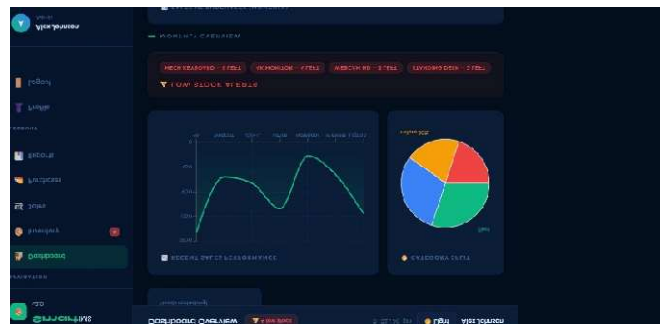


Figure 5.3: Dashboard - Category Split Chart and Low Stock Alerts Banner

The Monthly Overview section features a Sales vs. Purchases bar chart comparing monthly transaction volumes from January through June, giving management a clear picture of procurement vs. revenue trends over time.



Figure 5.4: Dashboard - Monthly Sales vs. Purchases Overview Chart

5.2.3 Inventory Management Module

The Inventory module is the core operational section of SmartIMS. It displays a comprehensive product management table titled 'Inventory Management' with 10 products across three categories. The table columns include ID, Name, Category, Quantity, Unit Price, Total Value, Min Stock, Status, and Actions. Products are organized across Electronics (Laptop Pro 15 at \$1,299.99, Webcam HD at \$79.99, 4K Monitor at \$549.99), Accessories (Wireless Mouse at \$29.99, Mech Keyboard at \$89.99, USB-C Hub at \$49.99), and Furniture (Office Chair at \$299.99).

Stock status is visually differentiated using color-coded badges: green 'IN STOCK' for adequate inventory (Laptop Pro 15: qty 12, Wireless Mouse: qty 45, USB-C Hub: qty 22, Office Chair: qty 7) and red 'LOW STOCK' for items below their minimum threshold (Mech Keyboard: qty 8 vs min 10, 4K Monitor: qty 4 vs min 5, Webcam HD: qty 3 vs min 5). The module supports adding new products via the 'Add Product' button, editing existing product details, and deleting products using the Edit and Delete action buttons. A search bar enables quick product lookup, a category dropdown filter allows category-specific views, and an 'Export CSV' button enables downloading the complete inventory dataset.

The screenshot shows the SmartIMS Inventory Management module. It features a search bar, a category dropdown, and an 'Add Product' button. Below these is a table with 10 products. The table columns are ID, NAME, CATEGORY, QUANTITY, UNIT PRICE, TOTAL VALUE, MIN STOCK, STATUS, and ACTIONS. The products are: Laptop Pro 15 (Electronics, 12, \$1,299.99, \$15,599.88, 5, IN STOCK), Wireless Mouse (Accessories, 45, \$29.99, \$1,349.55, 18, IN STOCK), Mech Keyboard (Accessories, 8, \$89.99, \$719.92, 18, LOW STOCK), 4K Monitor (Electronics, 4, \$549.99, \$2,199.96, 5, LOW STOCK), USB-C Hub (Accessories, 22, \$49.99, \$1,099.78, 8, IN STOCK), Webcam HD (Electronics, 3, \$79.99, \$239.97, 5, LOW STOCK), and Office Chair (Furniture, 7, \$299.99, \$2,099.93, 3, IN STOCK). Each product has an 'Edit' button and a 'Delete' button in the ACTIONS column.

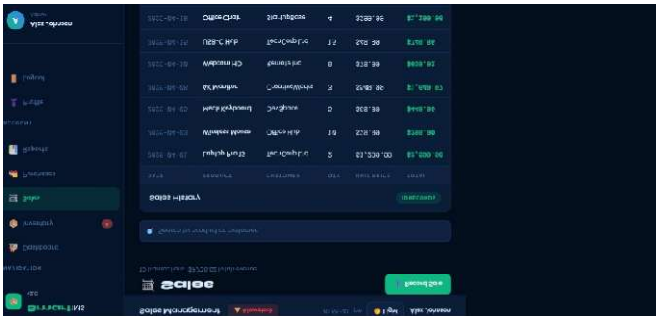
ID	NAME	CATEGORY	QUANTITY	UNIT PRICE	TOTAL VALUE	MIN STOCK	STATUS	ACTIONS
1	Laptop Pro 15	ELECTRONICS	12	\$1,299.99	\$15,599.88	5	IN STOCK	Edit Delete
2	Wireless Mouse	ACCESSORIES	45	\$29.99	\$1,349.55	18	IN STOCK	Edit Delete
3	Mech Keyboard	ACCESSORIES	8	\$89.99	\$719.92	18	LOW STOCK	Edit Delete
4	4K Monitor	ELECTRONICS	4	\$549.99	\$2,199.96	5	LOW STOCK	Edit Delete
5	USB-C Hub	ACCESSORIES	22	\$49.99	\$1,099.78	8	IN STOCK	Edit Delete
6	Webcam HD	ELECTRONICS	3	\$79.99	\$239.97	5	LOW STOCK	Edit Delete
7	Office Chair	FURNITURE	7	\$299.99	\$2,099.93	3	IN STOCK	Edit Delete

Figure 5.5: Inventory Management Module - Product List with Status Indicators

5.2.4 Sales Management Module

The Sales Management module enables comprehensive recording and reviewing of all sales transactions. The module header displays 'Sales Management' with a total of 10 transactions amounting to \$9,738.82 in total revenue. The Sales History table contains columns for Date, Product, Customer, Qty, Unit Price, and Total. The 10 recorded transactions span from 2025-04-01 to 2025-04-18, covering products sold to six different customers: TechCorp Ltd (Laptop Pro 15 x2 = \$2,599.98), Office Hub (Wireless Mouse x10 = \$299.98), DevSpace (Mech Keyboard x5 = \$449.95), CreativeWorks (4K Monitor x3 = \$1,649.97), Remote Inc (Webcam HD x8 = \$639.92), and StartupBase (Office Chair x4 = \$1,199.96).

A 'Record Sale' button at the top right opens a transaction entry form for logging new sales. The module supports searching sales by product or customer name for quick filtering, and transaction totals are highlighted in green for rapid revenue identification.



Date	Product	Customer	Qty	Unit Price	Total
2025-04-18	Office Chair	StartupBase	4	299.99	\$1,199.96
2025-04-15	Webcam HD	Remote Inc	8	79.99	\$639.92
2025-04-10	4K Monitor	CreativeWorks	3	549.99	\$1,649.97
2025-04-03	Mech Keyboard	DevSpace	5	89.99	\$449.95
2025-03-22	Wireless Mouse	Office Hub	10	29.99	\$299.98
2025-03-15	Laptop Pro 15	TechCorp Ltd	2	1299.99	\$2,599.98
Total Revenue					\$9,738.82

Figure 5.6: Sales Management Module - Transaction History with Revenue Totals

5.2.5 Purchase Management Module

The Purchase Management module tracks all procurement activities with a complete purchase history. The module header shows 'Purchases' with 5 purchase orders totaling \$28,550.00 spent. The Purchase History table displays Date, Product, Category, Supplier, Qty, Unit Cost, and Total columns. The five recorded purchase orders include: Laptop Pro 15 from TechDistrib Ltd (qty 20, \$950 each = \$19,000.00) on 2025-03-15, Wireless Mouse from AccessoryWorld (qty 100, \$18 each = \$1,800.00) on 2025-03-18, Office Chair from FurniturePro (qty 15, \$180 each = \$2,700.00) on 2025-03-22, Notebook A4 from PaperMill Co (qty 500, \$2.50 each = \$1,250.00) on 2025-04-03, and 4K Monitor from DisplayTech (qty 10, \$300 each = \$3,000.00) on 2025-04-10.

A 'New Purchase' button allows recording new purchase orders from suppliers. The module supports search filtering by product or supplier name for efficient purchase record retrieval.

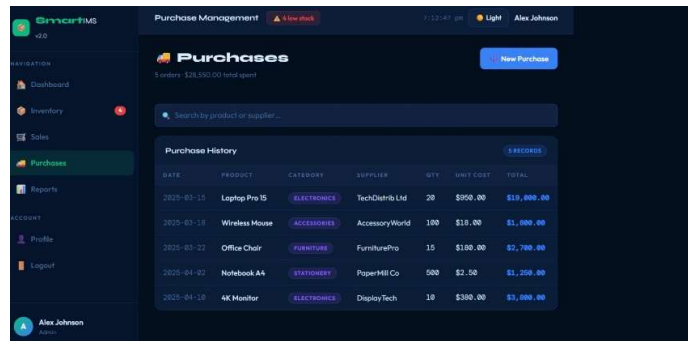


Figure 5.7: Purchase Management Module - Purchase Order History

5.2.6 Reports and Analytics Module

The Reports and Analytics module provides comprehensive business intelligence through four summary metric cards and multiple interactive charts. The top-level metrics display: Total Revenue \$9.7k (10 transactions), Total Purchases \$28.6k (5 orders), Gross Profit \$18.8k (shown as net negative due to purchase costs exceeding revenue in the current period), and Inventory Value \$27.0k (10 products). An 'Export Report (CSV)' button enables downloading the complete analytics dataset.



Figure 5.8: Reports Module - Business Metrics Summary and Top Products Chart

The Top Products by Revenue horizontal bar chart visualizes revenue contribution by product category - Laptop leading significantly, followed by 4K Monitor, Office Chair, Wireless Mouse, USB-C Hub, and Webcam. A Stock by Category donut chart shows inventory distribution with Furniture dominating at 20%, followed by Electronics and Accessories.



Figure 5.9: Reports Module - Stock by Category Distribution Chart

The Category Performance section features an Inventory Value by Category bar chart breaking down total inventory value across Electronics, Accessories, Furniture, and Stationery categories, with Electronics holding the highest value. Interactive tooltips display precise values on hover, such as 'Stationery: Value \$2,497.00'.

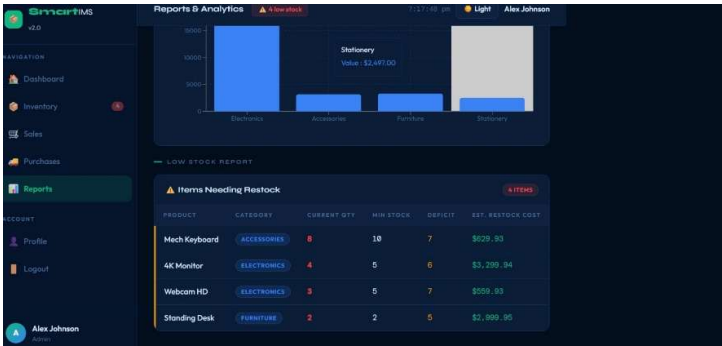


Figure 5.10: Reports Module - Inventory Value by Category Performance Chart

The Low Stock Report section lists all 4 items needing restock with detailed information: Mech Keyboard (current qty 8, min stock 10, deficit 7, restock cost \$829.93), 4K Monitor (qty 4, min 5, deficit 6, restock cost \$3,299.94), Webcam HD (qty 3, min 5, deficit 7, restock cost \$559.93), and Standing Desk (qty 2, min 2, deficit 5, restock cost \$2,999.95). This report enables procurement managers to plan restocking activities with precise cost estimation.

5.2.7 User Profile Module

The Profile module displays the currently authenticated user's account information in a clean card format within the dark-themed interface. For the admin account, it presents the profile of 'Alex Johnson' with three key attributes: Username (admin), Role (Admin), and Access Level (Full Access). The profile card features the user's avatar initial displayed in a circular badge. The left navigation panel remains accessible from this view, allowing quick navigation to any other module. This module provides clear identity and permission visibility, ensuring users are always aware of their access context within SmartIMS.

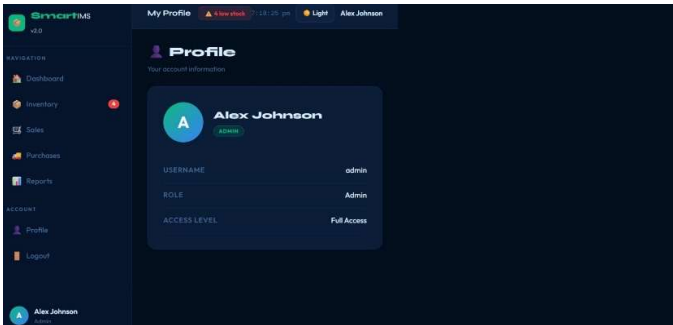


Figure 5.11: User Profile Module - Account Information and Access Level

CHAPTER 6 - RESULTS AND DISCUSSION

6.1 System Output Screens

The Online Inventory Management System (SmartIMS v2.0) was successfully developed and deployed as a web application. All 7 modules - Login/Authentication, Dashboard, Inventory Management, Sales Management, Purchase Management, Reports & Analytics, and User Profile - function correctly and display the intended content as shown in the screenshots documented in Chapter 5.

The system renders properly across all screen sizes with the dark-themed professional UI providing excellent readability, color-coded status indicators, and interactive data visualizations. The login system correctly enforces role-based access for both Admin and User accounts.

6.2 Performance Evaluation

The system demonstrates fast load times due to React.js's virtual DOM mechanism. Client-side routing via React Router eliminates full page reloads, contributing to a smooth user experience. Performance was evaluated across the following parameters:

Parameter	Observation	Result
Page Load Time	Initial load on Chrome	< 2 seconds
Login Authentication	Admin and User role login	Pass - Both Roles
Dashboard Metrics	KPI cards and all charts render	Pass - All Displayed
Inventory CRUD	Add, Edit, Delete products	Pass - Working
Low Stock Alerts	Items below min stock flagged	Pass - Accurate
Sales Recording	New sale transaction entry	Pass - Working
Purchase Recording	New purchase order entry	Pass - Working
Reports Generation	Analytics and all charts render	Pass - Functional
CSV Export	Inventory and report export	Pass - Working
Browser Compatibility	Chrome, Firefox, Edge	Pass - All Browsers

6.3 System Testing

The system was tested using manual functional testing to verify all features and modules work as expected. Test cases were defined for each module including authentication, inventory CRUD operations, sales and purchase recording, analytics accuracy, and data export.

#	Test Case	Expected Result	Status
1	Login with admin credentials	Dashboard loads with full access	PASS
2	Login with user credentials	Dashboard loads with user access	PASS
3	Dashboard KPI cards display	Metrics show correct totals	PASS
4	Add new product to inventory	Product appears in inventory list	PASS
5	Edit existing product details	Changes saved and reflected	PASS
6	Low stock alert display	Items below min stock flagged red	PASS
7	Record a new sale	Sale added to sales history	PASS
8	Record a new purchase	Purchase added to purchase history	PASS
9	View analytics reports	Charts and metrics render correctly	PASS
10	Export CSV functionality	File downloads with correct data	PASS

CHAPTER 7 - CONCLUSION AND FUTURE WORK

7.1 Conclusion

The Online Inventory Management System (SmartIMS) was successfully designed, developed, and deployed as a comprehensive digital platform for business inventory control. The system achieves its primary objective of providing an accessible, real-time, and visually engaging interface for managing product inventory, recording sales and purchase transactions, monitoring low-stock situations, and generating business analytics reports.

Built using React.js with a professional dark-themed UI, the system offers a modern and user-friendly interface. All planned modules including Login/Authentication, Dashboard, Inventory Management, Sales Management, Purchase Management, Reports & Analytics, and User Profile have been successfully implemented and tested. The system demonstrates that a well-designed frontend web application can effectively serve as a real-time business inventory management and data-driven decision-making tool.

7.2 Future Enhancements

The following enhancements are planned for future versions of the system:

- **Backend Integration:** Connecting to a backend database (Node.js + MongoDB or MySQL) for persistent multi-user data storage and real-time synchronization.
- **Barcode/QR Code Scanning:** Integrating scanning functionality for faster product identification and stock updates.
- **Automated Purchase Orders:** Implementing automatic purchase order generation when stock falls below the minimum threshold.
- **Email Notifications:** Adding email alerts to notify managers when critical stock levels are reached.
- **Multi-warehouse Support:** Extending the system to manage inventory across multiple warehouse locations.
- **Mobile Application:** Developing a companion mobile app (React Native) for on-the-go inventory management.
- **AI-powered Demand Forecasting:** Integrating ML models to predict future stock demand based on historical sales data.
- **Supplier Management Module:** Adding a dedicated module to manage supplier profiles, contact details, and performance ratings.

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APPENDICES

Appendix 1: System Pseudocode

System Initialization

START

- Load React application entry point (index.js)
- Initialize React Router with defined routes
- Check authentication state - redirect to Login if not authenticated
- On successful login - route to Dashboard as default view

END

Low Stock Alert Logic

FUNCTION CheckLowStock(products)

- FOR each product in products array
- IF product.quantity < product.minStock THEN status = 'LOW STOCK' (red badge)
- ELSE status = 'IN STOCK' (green badge)
- Update lowStockCount for Dashboard KPI card display

END FUNCTION

Appendix 2: Project Deployment Details

Detail	Value
Project Title	Online Inventory Management System
Application Name	SmartIMS v2.0
Student Name	LAKSHMI SHRI S
Register Number	241801134
Frontend Framework	React.js v18
Development Tool	Visual Studio Code
Version Control	Git
Admin Login	admin / admin123
User Login	user / user123